

## PROJECT AND TEAM INFORMATION

### Project Title

*ShrinkIt : Smart File Compression and Management System*

### Student / Team Information

|                                  |                                                                                                                                                                       |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Team Name:</i>                | <i>ShrinkIt</i>                                                                                                                                                       |
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## PROPOSAL DESCRIPTION

### Motivation

*We have noticed that digital data is increasing day by day, managing and storing files properly is becoming difficult as they are present in large numbers. Large files require more space and a lot of time in terms of transfer. Thus, saving disk space, transferring files conveniently, and managing files in bulk has become a big problem for users these days.*

*To solve this problem, we suggest making a project called ShrinkIt, by which the user would be able to compress their files. In this project, we have incorporated Huffman Coding Algorithm to reduce the size of a file without disturbing its actual data. The system will comprise features such as file compression/decompression, file type detection, algorithm recommendation, duplicate detection, file integrity check, and others. Moreover, this program will show the compression ratio, the saved space and time for compressing the file. We are going to use priority queues, binary trees, and hashing for this project. We also plan to use the concepts of OOPs to make the system well-structured.*

*The main idea of this project is to utilize and implement our knowledge of DSA and OOPs in building an effective and efficient file managing system.*

### State of the Art / Current solution

*Nowadays, we commonly use applications like WinRAR, WinZip, 7-Zip and Gzip for file compression. WinRAR is used for creating and extracting RAR/ZIP archives. 7-Zip is mainly used for high ratio compression of files and WinZip is used for compressing files and sharing files. Operating system also gives us features which already exist in ZIP option, which we can use for basic file management. It reduces size of the files, so that we can store and transfer them easily. However these applications mainly focus on compression and extraction and have limited file management. Therefore they do not show much detailed information about compression, like compression ratio, space saved and time taken.*

### Project Goals and Milestones

*ShrinkIt is a system which is easy to use, that compresses file size and also maintains data integrity. ShrinkIt provides file compression and extraction, duplicate file detection if a file is uploaded multiple times, and file management. It also shows details such as size of original file, compressed size, compression ratio and also space saved after compressing the file. It provides compression method which is most suitable for a file.*

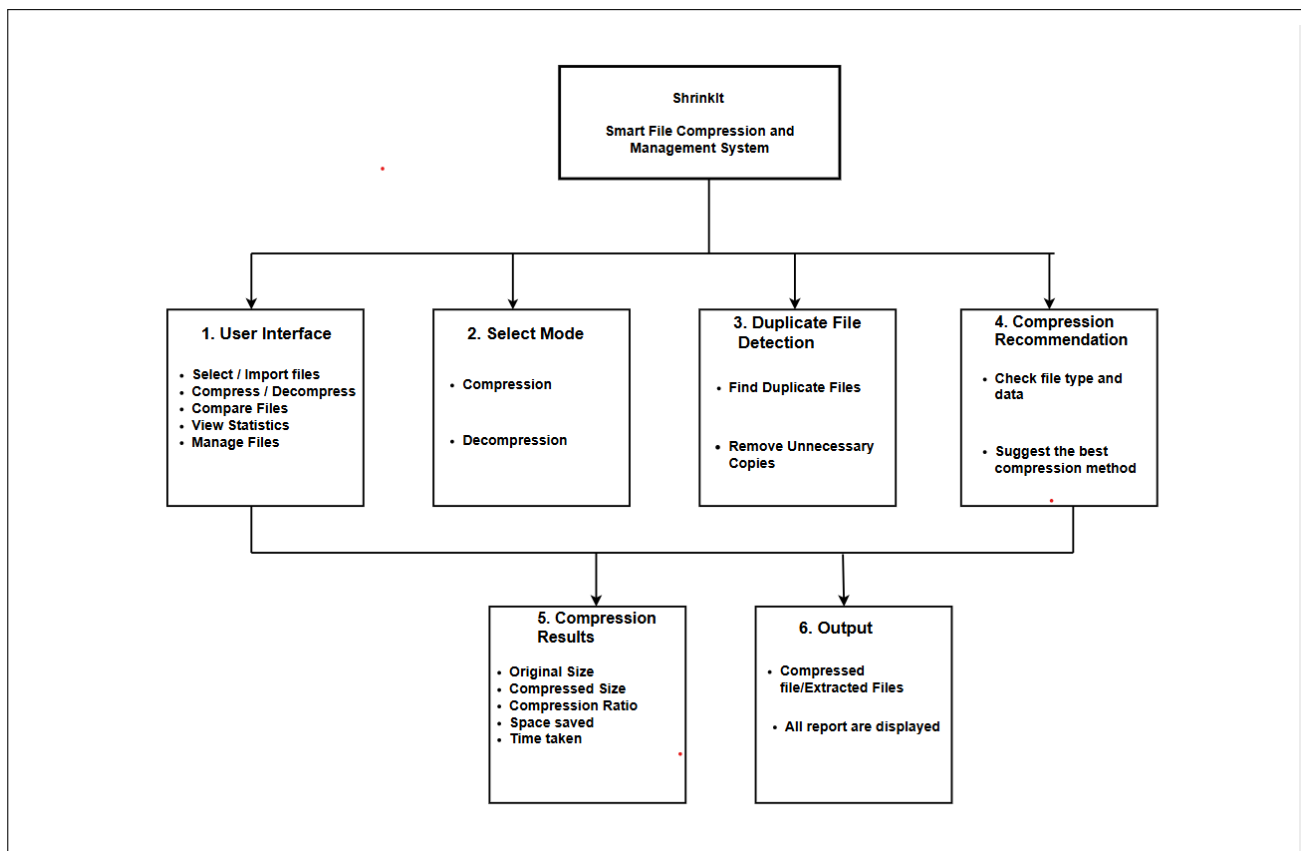
*Key Milestones:*

- 1. Research - It will study about the compression methods which exists already.*
- 2. Compression And Decompression - It will create a basic compression and decompression system that selects a file and make the changes according to users need.*
- 3. Duplicate Detection - Files which are repeated are automatically detected and same existing file is popped.*
- 4. Smart Features - Performs features so that the user can view, select and handle files.*
- 5. Compression Suggestion - Add and suggest the most appropriate compression method.*
- 6. Compression Details - Show compression statistics, compression size, compression ratio, space saved and time taken.*
- 7. It will test the files, bring all features together and create a user friendly interface.*

## Project Approach

*ShrinkIt is a project that aims to combine data structures, algorithms and OOP to make a functional file compression utility. Hashing is used for finding duplicate files, because it is faster than the direct content comparison. Arrays and strings stores and operates the file and text data, and the search and sort algorithms help to keep it ordered and find information. Here Trees , heaps , and priority queues are used in the compression algorithms to achieve required result. The algorithm used for compression is Huffman Encoding. The program reads the text file , for each character in it , counts all occurrences and builds a tree from them , using the min - heap structure . By Huffman's principle , the more common characters have the shorter codes , which decreases the total size of the file . The given text is stored separately , which allows extracting the original data. Aside from compression and extraction , the program finds duplicate files, compares files , and manages files with standard operations . It displays information about like the original size , compressed size , compression ratio , saved space , and the time spent during the operation.*

## System Architecture (High Level Diagram)



## Project Outcome / Deliverables

*ShrinkIt is proposed to be an efficient compression and file management system that uses Huffman Encoding to shrink files with no alterations to the original information. The software is supposed to include the following features: compression and decompression of files, detection of duplicate files, comparison of files, file management, and obtaining of compression statistics. Additionally, it will analyze the attributes of a text file and recommend the best method of compression instead of a default one.*

*The aim is to build an efficient file compression and management system which allow's the user to reduce the occupied space, remove the redundant data and measure the compression efficiency, which is represented in terms of statistics, like input size, output size, compression ratio, saved space, and time spent on the operation. The deliverables include ShrinkIt application, source code of the application written in C++, Huffman coding and decoding modules, duplicate, detection, comparison, and file management modules, compression statistics, test results, system architecture, and documentation.*

## Assumptions

*The project is assuming that the files input are readable, available and are not corrupted. The system will concentrate on text files where Huffman encoding is useful to compress files . The users must have access to storage and system resources for processing the files . The files after compression will contain all the necessary information to decode the Huffman code and reconstruct the original data .*

## References

- 1. C++ Learning Resources and Documentation*
- 2. GeeksforGeeks - DSA and Algorithms Resources*
- 3. Class Notes and Course Materials*
- 4. [www.google.com](http://www.google.com)*
- 5. ZIP File Format – PKWARE ZIP File Format Specification*